Introduction to Digital Twins

Portfolio

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1. Write the following systems of differential equations as systems of first order equations [10 points].

a) $\ddot{y}(t) = y(t)^2 - x(t) + e^t$
$\ddot{x}(t) = y(t) - x(t)^2 - e^t$

Solution: We introduce new variables. Let's define:

$x_1(t) = x(t)$, $x_2(t) = \dot{x}(t)$, $x_3(t) = y(t)$, $x_4(t) = \dot{y}(t)$

Taking derivatives with respect to time: we obtain

$$\begin{aligned}
\dot{x}_1(t) &= \dot{x}(t) = x_2(t) \\
\dot{x}_2(t) &= \ddot{x}(t) = y(t) - x(t)^2 - e^t \\
\dot{x}_3(t) &= \dot{y}(t) = x_4(t) \\
\dot{x}_4(t) &= \ddot{y}(t) = y(t)^2 - x(t) + e^t
\end{aligned}$$

Finally,

$$\begin{aligned}
\dot{x}_1(t) &= x_2(t) \\
\dot{x}_2(t) &= x_3(t) - x_1(t)^2 - e^t \\
\dot{x}_3(t) &= x_4(t) \\
\dot{x}_4(t) &= x_3(t)^2 - x_1(t) + e^t
\end{aligned}$$

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b) $\ddot{x}(t) = 2y(t) - 4t^2 x(t)$
$\ddot{y}(t) = -2x(t) - 2t\dot{x}(t)$

Solution: We introduce new variables. Let's define:

$x_1(t) = x(t)$, $x_2(t) = \dot{x}(t)$, $x_3(t) = y(t)$, $x_4(t) = \dot{y}(t)$

Now, we can rewrite the given system of equations in terms of these new variables:

$$\begin{aligned}
\dot{x}_1(t) &= \dot{x}(t) = x_2(t) \\
\dot{x}_2(t) &= \ddot{x}(t) = 2x_3(t) - 4t^2 x_1(t) \\
\dot{x}_3(t) &= \dot{y}(t) = x_4(t) \\
\dot{x}_4(t) &= \ddot{y}(t) = -2x_1(t) - 2tx_2(t)
\end{aligned}$$

So, the system of first-order equations is:

$$\begin{aligned}
\dot{x1}(t) &= x_2(t) \\
\dot{x2}(t) &= 2x_3(t) - 4t^2 x_1(t) \\
\dot{x3}(t) &= x_4(t) \\
\dot{x4}(t) &= -2x_1(t) - 2tx_2(t)
\end{aligned}$$

The system can also be written in the form of dx/dt = Ax + Bu. These equations can be written in matrix form as:

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -4t^2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & -2t & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

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c) $\ddot{x}(t) = \ddot{y}(t)x(t)$
$\ddot{x}(t) = y(t) - x(t)^2 - e^t$

Solution: We introduce new variables. Let's define:

$x_1(t) = x(t)$, $x_2(t) = \dot{x}(t)$, $x_3(t) = y(t)$, $x_4(t) = \dot{y}(t)$

Taking derivatives with respect to time: we obtain

$$\begin{aligned} \dot{x}_1(t) &= \dot{x}(t) = x_2(t) \\ \dot{x}_2(t) &= \ddot{x}(t) = x_4(t)x_1(t) - x_4(t)^2 x_1(t) \\ \dot{x}_3(t) &= \dot{y}(t) = x_4(t) \\ \dot{x}_4(t) &= \ddot{y}(t) = -x_3(t) \end{aligned}$$

Finally, the system of first-order equations is:

$$\begin{aligned} \dot{x1}(t) &= x_2(t) \\ \dot{x2}(t) &= x_4(t)x_1(t) - x_4(t)^2 x_1(t) \\ \dot{x3}(t) &= x_4(t) \\ \dot{x4}(t) &= -x_3(t) \end{aligned}$$

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d) $\dfrac{d^3y}{dt^3} = \dddot{y}(t)x(t)^2 - 3\dot{y}(t)^2 x(t)$
$\ddot{y}(t) = t + \dot{x}(t)$

Solution: We introduce new variables. Let's define:

$x_1(t) = y(t)$, $x_2(t) = \dot{y}(t)$, $x_3(t) = \ddot{y}(t)$, $x_4(t) = x(t)$
Now, we can rewrite the given system of equations in terms of these new variables:

$$\begin{aligned} \dot{x}_1(t) &= \dot{y}(t) = x_2(t) \\ \dot{x}_2(t) &= \ddot{y}(t) = x_3(t) = t + \dot{x_4}(t) \\ \dot{x}_3(t) &= \dddot{y}(t) = x_3(t)x_4(t)^2 - 3x_2(t)^2 x_4(t) \\ \dot{x}_4(t) &= \dot{x}(t) = x_3(t) - t \end{aligned}$$

Finally, the system of first-order equations is:

$$\dot{x1}(t) = x_2(t)$$
$$\dot{x2}(t) = t + \dot{x}_4(t)$$
$$\dot{x3}(t) = x_3(t)x_4(t)^2 - 3x_2(t)^2x_4(t)$$
$$\dot{x4}(t) = x_3(t) - t$$

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2. Write the following differential equations as systems of first order equations [10 points].

a) $\ddot{x}(t) + t^2\dot{x}(t) + 3x(t) = 0$
b) $\ddot{x}(t) = -2x(t) - 10\dot{x}(t)$
c) $\ddot{y}(t) = 2\sqrt{e^{2t} - y(t)^2}$
d) $2\ddot{x}(t) - 5\dot{x}(t) + x(t) = 0$

Solution-2a: We introduce new variables. Let's define:

$x_1(t) = x(t)$, $x_2(t) = \dot{x}(t)$

Taking derivatives with respect to time: we obtain

$$\dot{x}_1(t) = \dot{x}(t) = x_2(t)$$
$$\dot{x}_2(t) = \ddot{x}(t) = -t^2x_2(t) - 3x_1(t)$$

So finally,

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -3x_1(t) - t^2x_2(t)$$

The system can also be written in the form of dx/dt = Ax + Bu.

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -3 & -t^2 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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Solution-2b: We introduce new variables. Let's define:

$x_1(t) = x(t)$, $x_2(t) = \dot{x}(t)$

Taking derivatives with respect to time: we obtain

$$\dot{x}_1(t) = \dot{x}(t) = x_2(t)$$
$$\dot{x}_2(t) = \ddot{x}(t) = -2x_1(t) - 10x_2(t)$$

So finally,

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -2x_1(t) - 10x_2(t)$$

The system can also be written in the form of dx/dt = Ax + Bu.

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -10 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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Solution-2c: We introduce new variables. Let's define:

$y_1(t) = y(t)$ and $y_2(t) = \dot{y}(t)$

Taking derivatives with respect to time: we obtain

$$\dot{y}_1(t) = \dot{y}(t) = y_2(t)$$
$$\dot{y}_2(t) = \ddot{y}(t) = 2\sqrt{e^{2t} - y_1(t)^2}$$

So finally,

$$\dot{y}_1(t) = y_2(t)$$
$$\dot{y}_2(t) = 2\sqrt{e^{2t} - y_1(t)^2}$$

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Solution-2d: We introduce new variables. Let's define:

$x_1(t) = x(t)$, $x_2(t) = \dot{x}(t)$

Taking derivatives with respect to time: we obtain

$$\dot{x}_1(t) = \dot{x}(t) = x_2(t)$$
$$\dot{x}_2(t) = \ddot{x}(t) = -\frac{1}{2}x_1(t) + \frac{5}{2}x_2(t)$$

So finally,

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -\frac{1}{2}x_1(t) + \frac{5}{2}x_2(t)$$

The system can also be written in the form of dx/dt = Ax + Bu.

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1/2 & 5/2 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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3. Solve equations (b) and (d) in 2) numerically using the Euler's and Runge-Kutta's method [10 points].

Solution: Matlab Code File : PF-3b.m & PF-3d.m in folder "Question-3b" & in "Question-3d"

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4. Solve the following system on the interval [0,2] using Euler and Runge-Kutta method [10 points].

$$\ddot{x}(t) = 2y(t) - \sin(4t^2 x(t)), x(0) = 1, \dot{x}(0) = 2$$
$$\ddot{y}(t) = -2x(t) - \frac{1}{2t^2\dot{x}(t)^2 + 3}, y(0) = 1, \dot{y}(0) = 0$$

Solution: Matlab Code File : PF-4.m in folder "Question-4"

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5. Make a simulation of a digital twin for autonomous ship that travel from A to B in Unity 3D. The simulation will be evaluated based on the following criteria:

a) The presentation of mathematical modelling of the ship [10 points]. The state generated by the model will serve as the state of the physical ship.

Solution: Matlab Code File : PF-5a.m in folder "Question-5a"

b) The presentation of the simulation and visualization in the Unity 3D [10 points].

Solution: "Digital Twin-Question-5c" Unity File

c) The presentation of state and parameter estimation of the digital model in Unity 3D. The estimation algorithm should be based on the following paper: A. Hasan, M. Tahavori, and H. Midtiby, Model-based fault diagnosis algorithms for robotic systems, IEEE Access, vol. 11, pp. 2250-2258, 2023 [30 points].

Solution: Matlab Code File : PF-5c.m in folder "Question-5c & " Digital Twin-Question-5c" Unity File

d) A video demonstration of the autonomous ship that goes from A to B [10 points].

Solution: "Autonomous-Ship-Digital-Twin.mp4" file in the folder "Digital Twin - Portfolio"